newspaper. Expect that raw computational speed will more than double each two years. Learning about the current state of computer power is simply not enough; you need to understand and track how it is evolving.

Keeping this in mind, we can jump into an overview of how execution speed is limited by computer hardware. Since computers are composed of many subsystems, the time required to execute a particular task will depend on two primary factors: (1) the speed of the individual subsystems, and (2) the time it takes to transfer data between these blocks. Figure 4-5 shows a simplified diagram of the most important speed limiting components in a typical personnel computer. The **Central Processing Unit (CPU)** is the heart of the system. As previously described, it consists of a dozen or so registers, each capable of holding 32 bits (in present generation personnel computers). Also included in the CPU is the digital electronics needed for rudimentary operations, such as moving bits around and fixed point arithmetic.

More involved mathematics is handled by transferring the data to a special hardware circuit called a **math coprocessor** (also called an **arithmetic logic unit**, or **ALU**). The math coprocessor may be contained in the same chip as the CPU, or it may be a separate electronic device. For example, the addition of two floating point numbers would require the CPU to transfer 8 bytes (4 for each number) to the math coprocessor, and several bytes that describe what to do with the data. After a short computational time, the math coprocessor would pass four bytes back to the CPU, containing the floating point number that is the sum. The most inexpensive computer systems don't have a math coprocessor, or provide it only as an option. For example, the 80486DX microprocessor has an internal math coprocessor, while the 80486SX does not. These lower performance systems replace *hardware* with *software*. Each of the mathematical functions is broken into

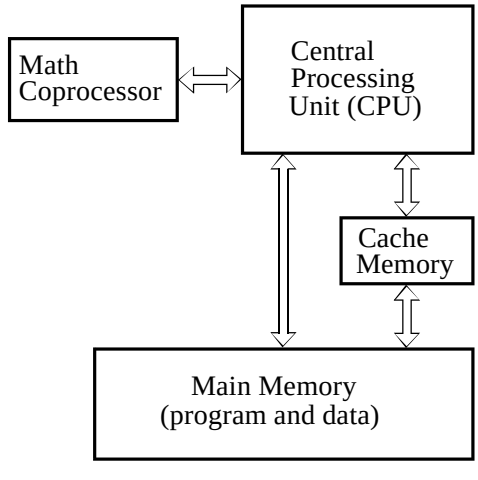

FIGURE 4-5
Architecture of a typical computer system. The computational speed is limited by: (1) the speed of the individual subsystems, and (2) the rate at which data can be transferred between these subsystems.